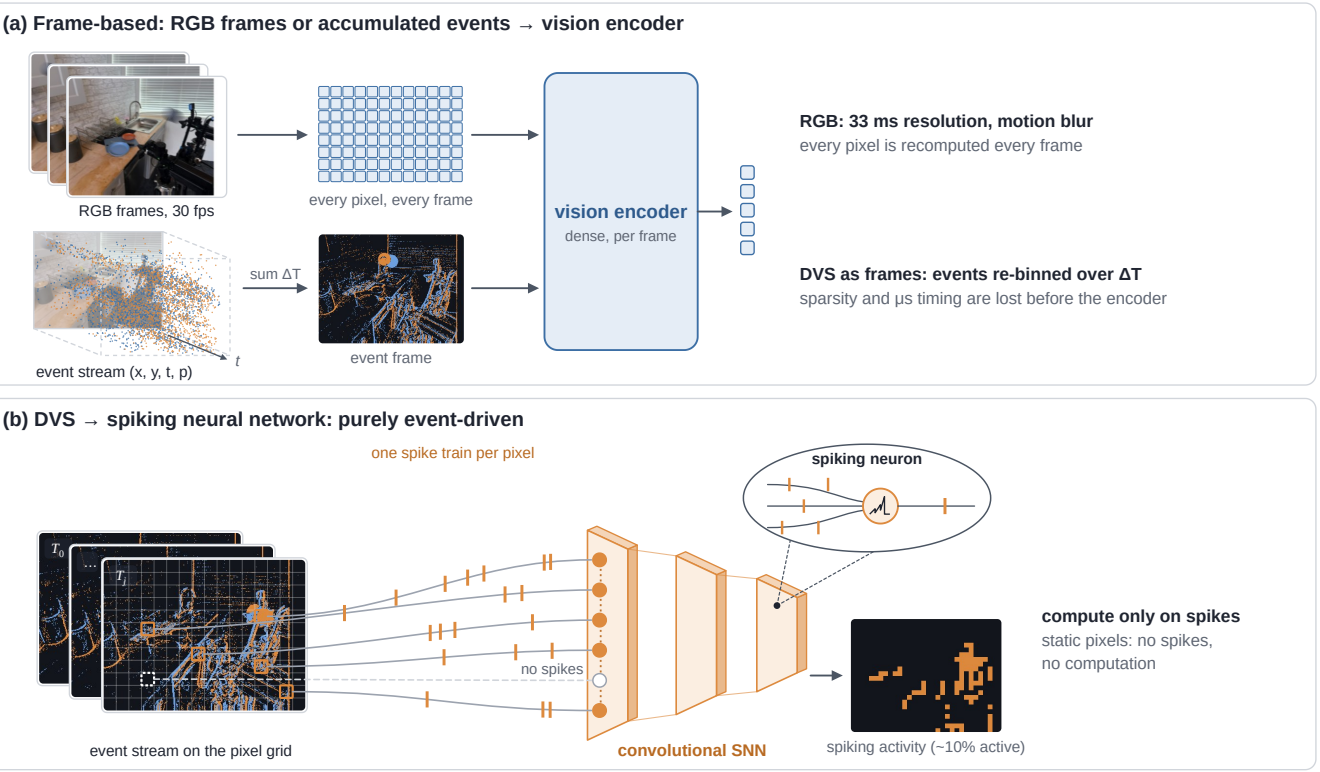

(a) Frame-based: RGB frames or accumulated events → vision encoder
RGB frames, 30 fps
every pixel, every frame
vision encoder
dense, per frame
RGB: 33 ms resolution, motion blur
every pixel is recomputed every frame
sum ΔT
event stream (x, y, t, p)
t
event frame
DVS as frames: events re-binned over ΔT
sparsity and µs timing are lost before the encoder
(b) DVS → spiking neural network: purely event-driven
one spike train per pixel
spiking neuron
T_0
T_j
no spikes
event stream on the pixel grid
convolutional SNN
spiking activity (~10% active)
compute only on spikes
static pixels: no spikes,
no computation